

Panel MS-AR(1): random intercepts with $E[z] = 0$

Seasonally adjusted real GDP per worker

1 Model

One specification. Joint panel Markov-switching AR(1) around a common linear trend, with Normal random intercepts. Outcome is log seasonally adjusted real GDP per worker (2026-09-17 vintage), full sample.

$$y_{it} = a_i + g t + z_{it}, \tag{1}$$

$$a_i \sim \mathcal{N}(\alpha, \omega^2), \tag{2}$$

$$z_{i,t+1} = (1 - \rho(s_{it}))\mu(s_{it}) + \rho(s_{it}) z_{it} + \sigma(s_{it}) \varepsilon_{it}, \tag{3}$$

$$s_{i,t+1} \sim \Pi(\cdot \mid s_{it}). \tag{4}$$

Three regimes. σ and ρ switch. $|\rho| < 0.99$. The intercept and the regime means are not separately identified, so the unconditional mean of the cycle is restricted to $E[z] = 0$ (the median regime mean is *not* pinned at 0). Random intercepts use 11-point Gauss–Hermite. Plotted cycles use the posterior mean of a_i . Standard errors are delta-method from a numerical Hessian.

2 Random intercepts, common linear trend, $E[z]=0$

2026-09-17 SA panel, full sample, $y_{it} = a_i + gt + z_{it}$, $E[z] = 0$. 74 countries, 8670 observations, 1950-Q1–2026-Q2. Fitted countries: 74. Observations: 8670. Log-likelihood: 23496.84. Random intercepts $a_i \sim N(\alpha, \omega^2)$ with $\alpha = -5.1672$ (0.0529), $\omega = 0.5041$ (0.0058). Posterior-mean a_i : mean -5.220, min -7.782, max -3.640. Common slope $g = 0.0055$ (0.0007).

Table 1: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-1.0703 (0.0963)	-0.0794 —	0.2903 (0.0233)
σ	0.0738 (0.0025)	0.0186 (0.0005)	0.0069 (0.0001)
ρ	0.9899 (0.0005)	0.9900 (0.0001)	0.9899 (0.0001)
Π_0	0.6943 (0.0323)	0.2876 (0.0329)	0.0181 (0.0083)
Π_1	0.0638 (0.0079)	0.8778 (0.0109)	0.0584 (0.0068)
Π_2	0.0052 (0.0033)	0.0506 (0.0067)	0.9442 (0.0055)
π	0.0971 (0.0100)	0.4262 (0.0209)	0.4767 (0.0248)

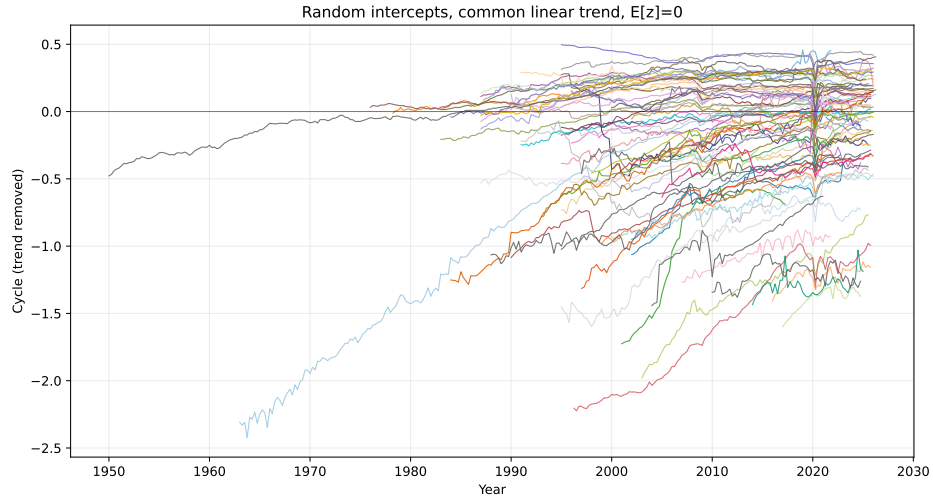


Figure 1: Country cycles after removing the posterior-mean intercept and the common linear trend.